

Semantic Fiber Bundle: Bio-Inspired Geometric Axioms and Theorems for Structured Large Language Models

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Abstract

*The architectural evolution of large language models (LLMs)—from Dense Transformer to Mixture-of-Experts (MoE), from sparse attention to model editing—has unfolded as a sequence of engineering fragments lacking a unified mathematical language. Inspired by four independent yet isomorphic natural systems—celestial coordinate frameworks, neural encoding, immune affinity maturation, and embryonic metamerism—we propose the **Semantic Fiber Bundle (SFB)**: an axiomatic, derivable, and computable geometric paradigm.*

*We construct the semantic manifold $\widetilde{M}$ rigorously, propose five axioms (including the lattice-connection axiom), and establish three core constructions: (1) **Semantic Place-Field Construction**—routing completeness with soft-tuned ambiguity control; (2) **Precession-Affinity Maturation**—curvature-governed knowledge drift and iterative correction; (3) **Gravitational Attention**—attention kernel as symplectic generating kernel (geometric heuristic framework). Complementing the spatial fiber-bundle structure, **persistent homology** is introduced as the temporal pillar, defining the **Dual-Track Epoch** (innate vs. adaptive knowledge lifespan).*

*Numerical validation on Qwen1.5-MoE-A2.7B (14.3B parameters, 60 experts) demonstrates: (i) expert count N exerts significant control on ambiguity-set measure (overlap drops from 89.07% with 4 experts to **13.31%** with 60 experts at threshold 0.9, approximate power-law decay); (ii) gate perturbation induces an **S-shaped threshold effect**—low-noise plateau $\sim 42\%$, mid-noise steep rise to $\sim 54\%$, high-noise saturation $\sim 55\%$ —revealing a binary dichotomy of knowledge types (immune vs. vulnerable); (iii) cross-domain persistent homology reveals science texts achieve the highest lifespan in both connected components ($PH_0=49.86$) and topological loops ($PH_1=3.00$), whereas mathematical theorems exhibit "crystallized" isolation due to cross-branch semantic fracture ($PH_0=38.68$, $PH_1=2.02$); (iv) the RoPE-distortion experiment confirms that semantic place-field concentration decays strictly with token distance (Short $CV=2.49 \rightarrow$ Long $CV=1.06 \rightarrow$ Ultra $CV=0.66$, **73.3%** total collapse, reflecting gate distribution degradation with token distance), providing empirical evidence for the compensatory mechanism of Lattice-Connection Axiom 3'.*

Keywords: Mixture-of-Experts; fiber bundle; symplectic geometry; persistent homology; semantic manifold; knowledge editing; RoPE distortion

1. Introduction: From Astronomical Observation to Biological Evolution

The architectural evolution of large language models—MoE routing, RoPE position encoding, and model editing—has proceeded independently, lacking a unified mathematical language to relate them as different projections of the same geometric object. This fragmentation leads to three core challenges: (1) routing lacks flexible ambiguity control; (2) knowledge updating lacks a quantitative upper bound on drift; (3) attention mechanisms lack symplectic geometric regularity constraints. The Semantic Fiber Bundle framework proposed in this paper provides a unified axiomatic answer to these three problems.

How do complex systems establish order amidst chaos? Nature provides four independent yet isomorphic answers:

- **Neural System:** Hippocampal place cells encode continuous physical space into discrete "place fields", with underlying grid cells providing hexagonal lattice coordinates;
- **Immune System:** Antibodies undergo somatic hypermutation within a fixed framework for local affinity maturation, achieving iterative knowledge updates;
- **Developmental System:** Hox genes drive three-dimensional embryonic differentiation into discrete "metameres" through a one-dimensional collinear expression gradient on chromosomes;
- **Astronomical System:** Ancient astronomers divided ecliptic-equatorial stars into coordinate grids to precisely mark the apparent positions of the Sun, Moon, and planets.

These four systems share the same mathematical structure: **on a continuous substrate, a covering is formed by discrete, functionally specialized modules; modules communicate locally; modules undergo local updates over time; global stability emerges from local dynamic corrections.**

The core contribution of this paper is not to propose new engineering components, but to **provide a set of bio-inspired geometric axioms and derivable theorems for existing and future structured LLM architectures.**

1.1 Four-fold Inspiration Mapping

Biological/Natural System	Mathematical Structure	Paper Correspondence
Hippocampal Place Cells	Semantic Metamere Cover	Theorem 1 (Semantic Place-Field)
Hox Gene Collinearity	Semantic Hox Gradient	Proposition 2.5 (Metamerization Proposition)
Immune Affinity Maturation	Curvature Control	Theorem 2 (Precession-Affinity Maturation)
Celestial Coordinate Grid	Lattice Connection	Axiom 3' (Lattice Connection Axiom)

1.2 Key Contributions

- Propose the Semantic Fiber Bundle axiomatic framework, unifying the geometric language of MoE routing, attention, and knowledge editing;
- Demonstrate significant control of expert count on ambiguity set (overlap drops to 13.31% with 60 experts);
- Discover the "immune/vulnerable group" binary dichotomy of knowledge types and the S-shaped threshold effect of gating networks;
- Reveal the "crystallization" topological isolation phenomenon of mathematical theorems (PH_0/PH_1 double bottom);
- Verify that long-text RoPE distortion causes a 73.3% collapse in semantic place-field concentration.

2. Related Work

2.1 MoE Architecture: From Engineering to Geometry

Switch Transformer [1] first scaled MoE to trillion-parameter levels, yet its routing mechanism remains at the heuristic level—the load-balancing loss was introduced to alleviate training instability rather than from an understanding of semantic space structure. Our Semantic Fiber Bundle framework redefines experts as a discrete cover over the semantic manifold $\widetilde{M}$, elevating the routing decision from "which expert to select" to "into which place-field does the token fall on the manifold," transforming load balancing from a tuning trick into **the geometric inevitability of lattice connections**.

MegaBlocks [2] optimized MoE GPU utilization at the systems level, but its block-sparse structure is still viewed as a linear algebra implementation detail. We point out that this block sparsity is precisely the hardware-level projection of **Lattice Connection Axiom 3'**: hexagonal lattice symmetry naturally induces block-sparse adjacency matrices, reducing inter-mansion communication complexity from $O(n^2)$ to $O(n)$.

2.2 Model Editing: From Point Surgery to Gauge Transformation

ROME [3] achieved precise intra-layer editing of individual facts, using Rank-One matrix updates to modify specific facts with surgical precision. However, its editing operations lack theoretical constraints on "chain reactions." Our **Precession-Affinity Maturation construction(Theorem 2)** reinterprets model editing as a gauge transformation of the connection form: under bounded curvature, the drift from local editing is controlled by $\frac{1}{1-\delta} \int_{U_i} \|R^\nabla\|_{op} dVol_g$ extending ROME's "point surgery" to "controlled neighborhood updating."

MEMIT [4] extended ROME from single-point to batch editing, yet still treats knowledge as independent fact entries. Our **Dual-Track Epoch(Definition 5.1)** reveals topological differences between knowledge types: innate knowledge (mathematical theorems, physical constants) corresponds to $d_\alpha = \infty$ barcodes and should not be edited; adaptive knowledge (legal provisions, policies) corresponds to finite-lifespan barcodes amenable to iterative gauge transformation updates.

2.3 Agent Architecture: From Prompt Engineering to Cognitive Geometry

Frameworks such as OpenAI Agents SDK [7] and AutoGPT [8] encapsulate LLMs as agents capable of autonomous decision-making, but their tool invocation and planning logic remain at the prompt engineering level, lacking formal descriptions of "how knowledge is stored, how it evolves, and how it is forgotten." Our Semantic Fiber Bundle framework provides agents with **an axiomatizable cognitive geometry**: the Guard mechanism corresponds to curvature monitoring, the BackgroundCurator to precession-affinity maturation, and the ethics engine to gauge transformation constraints on connections.

3. Constructive Definition of Semantic Manifold and Metamerization

3.1 Semantic Manifold

Definition 2.1 (Semantic Manifold) Let $\mathcal{D} = \{x^{(n)}\}_{n=1}^N \subset \mathcal{X}$ be the pre-training corpus and $\phi : \mathcal{X} \rightarrow \mathbb{R}^d$ be the pre-trained encoder. Define the semantic point set $\mathcal{M}_0 = \phi(\mathcal{D}) \subset \mathbb{R}^d$, and take its closure in the subspace topology of $\mathbb{R}^d$ as $M := \overline{\mathcal{M}_0}$. Note that M is a closure and may not satisfy manifold conditions; only after Construction 2.2's local PCA denoising and thin-plate spline smoothing does $\widetilde{M}$ become a smooth embedded submanifold. For narrative convenience, we refer to M as the **semantic manifold** throughout this paper.

Construction 2.2 (Smoothing) Through local PCA denoising, spectral clustering smoothing, and thin-plate spline interpolation, we construct a C^∞ smooth embedded submanifold $\widetilde{M} \subset \mathbb{R}^d$. Dimension estimation uses the TwoNN method; empirically $m \ll d$ (e.g., $d = 2048, m \approx 32$).

3.2 Semantic Hox and Metamerization

In embryonic development, Hox genes drive three-dimensional metameric differentiation through a collinear expression gradient on a one-dimensional sequence. This principle maps to semantic space: **semantic metameres are not artificially pre-defined bins, but the result of self-organized differentiation of a continuous semantic manifold under a "Semantic Hox" gradient.**

Proposition 2.5 (Metamerization Theorem) Driven by the Hox gradient, the initially homogeneous parameter space Θ converges through training to discrete, functionally specialized **semantic metameres** $\{U_i\}_{i=1}^N$, with adjacent metameres simultaneously maintaining topological adjacency in both the semantic manifold M and the parameter space Θ (dual topology preservation).

4. Axiom System

Let the semantic space be the base manifold M (compact Riemannian manifold, metric g), the total model parameter space be the bundle E , with projection $\pi : E \rightarrow M$. The fiber $F_x = \pi^{-1}(x)$ is the activatable parameter subspace at semantic point x .

Axiom 1 (Routing Axiom) For any input $x \in M$, there exists a finite open cover $\mathcal{U} = \{U_i\}_{i=1}^N$ such that $x \in \bigcup_{i=1}^N U_i$.

Axiom 2 (Locality Axiom) Within U_i , model behavior depends only on the local fiber F_i and is independent of global parameters.

Axiom 3 (Connection Axiom) Inter-mansion communication is accomplished through parallel transport $P_\gamma : F_{\gamma(0)} \rightarrow F_{\gamma(1)}$ defined by the connection $\nabla : \Gamma(E) \rightarrow \Omega^1(M, E)$; direct coupling is forbidden.

Axiom 3' (Lattice Connection) Under local trivialization, the Christoffel symbols Γ_{ij}^k of the connection ∇ possess **hexagonal lattice symmetry**, allowing parallel transport along lattice paths to decompose into products of discrete translation operators, reducing computational complexity from $O(n^2)$ to $O(n)$.

Axiom 4 (Curvature Axiom) $R^\nabla \neq 0$ implies that parallel transport depends on the path. $\rightarrow$ Semantically: the same token reaches different experts in different contexts $\rightarrow$ Behaviorally: the same question receives different answers $\rightarrow$ This is the geometric definition of "hallucination." In MoE systems, the root cause of non-zero curvature is the nonlinear response of the gating network (Router) to positional encoding (RoPE)—distant tokens' gate logits are distorted by RoPE in long texts, causing routing decisions to deviate from the shortest geodesic path. Curvature R^∇ quantifies the upper bound of semantic drift.

Axiom 5 (Precession Axiom) Knowledge updating is a gauge transformation of the connection form. Under local trivialization, we adopt the Abelian approximation (U(1) connection): $A \rightarrow A + d\alpha$. This approximation applies to local rank-one editing operations such as ROME/MEMIT; the full non-Abelian generalization ($A \rightarrow g^{-1}Ag + g^{-1}dg$) is left for future work.

5. Core Theorems and Proofs

5.1 Theorem 1: Semantic Place-Field Construction

Definition 4.1 (Semantic Place-Field) For each mansion U_i , define its **place-field function**

$$\rho_i(x) = \exp\left(-\frac{d_g(x, c_i)^2}{2\sigma_i^2}\right) / Z(x),$$

where c_i is the reference star (prototype vector), σ_i is the mansion radius, and $Z(x)$ is the competitive inhibition normalization factor. Routing is no longer a Boolean decision, but an **activation rate vector** $\vec{\rho}(x) \in \Delta^{N-1}$ (probability simplex).

Competitive Inhibition Lemma When the reference star distance of two mansions U_i, U_j satisfies $d_g(c_i, c_j) < \sigma_i + \sigma_j$, competitive inhibition occurs at the boundary: $\partial\rho_i/\partial d_g(c_i, c_j) < 0$.

Theorem 1 (Semantic Place-Field Construction) Let M be a compact Riemannian manifold and $\{U_i\}_{i=1}^N$ be semantic metamerers. Then there exist smooth local sections $\sigma_i \in \Gamma(U_i, E)$ and place-field functions ρ_i such that the global section $\sigma(x) = \sum_i \rho_i(x)\sigma_i(x)$ satisfies complete coverage, soft-tuned ambiguity, and semantic consistency preservation (Lipschitz continuity bound).

5.2 Theorem 2: Precession-Affinity Maturation

Assumption 2.1 (Small-Loop Assumption) All temporal closed loops γ_t satisfy $\|\oint_{\gamma_t} A\|_{op} \leq \delta < 1$.

Theorem 2 (Precession Conservation) Let ∇ be a connection on E , with curvature $R^\nabla \in \Omega^2(M, \text{End}(E))$, and γ_t be a temporal closed loop. Then:

1. **Flat conservation:** If $R^\nabla \equiv 0$, then $\text{Hol}(\nabla, \gamma_t) = \text{Id}$;
2. **Curvature control:** If $R^\nabla \neq 0$ within some mansion U_i , then under Assumption 2.1

$$\| \text{Hol}(\nabla, \gamma_t) - \text{Id} \|_{op} \leq \frac{1}{1 - \delta} \int_{U_i} \| R^\nabla \|_{op} d\text{Vol}_g.$$

Remark 4.2 (Computability of Curvature) In practical systems, $\| R^\nabla \|_{op}$ can be approximated by the Frobenius norm change of parameters before and after editing, serving as a proxy indicator for curvature drift. This proxy maintains a monotonic relationship with the true Hessian trace under local linearization. In experiments, the curvature proxy reaches a maximum of 0.87 (normalized) at gate perturbation $\varepsilon=0.9$, far below 1, validating the reasonableness of the small-loop assumption ($\delta < 1$) in model editing scenarios.

5.3 Construction 3: Symplectic Geometric Heuristic Framework for Gravitational Attention

Construction 3.1 (Embedding to Cotangent Bundle Mapping) Let $q \in M$ be the Query embedding and $p \in \mathbb{R}^d$ be the Key embedding. Define the mapping $\Phi : M \times \mathbb{R}^d \rightarrow T^*M$, where $g_q(W_{Kp}, \cdot) \in T_q^*M$ is the musical isomorphism. The Jacobian of Φ has full rank, hence it is a local diffeomorphism.

Proposition 3.1 (Symplectic Geometric Correspondence) Let (M, g) be a semantic Riemannian manifold, with T^*M equipped with the canonical symplectic form ω . For an input sequence $X \subset M$, define the geodesic attention kernel

$$A_{ij} = \frac{\exp\left(-\lambda d_g(x_i, x_j)^2/2\right)}{Z_i}.$$

Then A_{ij} satisfies three necessary conditions for a symplectic generating kernel—Gaussian decay ($A_{ij} \leq \exp(-\lambda d_g^2/2)$), measure-preserving normalization ($\sum_j A_{ij} = 1$), and locality ($\lim_{d_g \rightarrow \infty} A_{ij} = 0$)—whose formal structure is compatible with the Type-I generating function $S(q, \tilde{p})$. Numerical verification (Experiment 3) confirms that all three properties hold under random initialization. **Note:** The momentum projection $\tilde{p}$ of the complete symplectic map $(\tilde{q}, \tilde{p}) = \varphi(q, p)$ requires completing the momentum update rule (involving implicit integration of velocity-position coupling), which is left for future work.

6. Persistent Homology: The Topological Pillar of Temporal Evolution

The fiber bundle (M, E, π, ∇) characterizes the **spatial structure**. The **temporal evolution** of knowledge requires an independent mathematical pillar—**Persistent Homology**. The two complement each other: the fiber bundle governs "which mansion does it fall into right now," while persistent homology governs "how long can this mansion survive."

Definition 5.1 (Dual-Track Epoch) Let $M_\varepsilon = \bigcup_{x \in X} B_g(x, \varepsilon)$ be the ε -thickening of the semantic point cloud, with persistent homology barcodes of $PH_k(M_\varepsilon)$ given by $\{[b_\alpha, d_\alpha)\}$. Each barcode is labeled as:

- **Innate:** d_α does not die within the maximum experimental observation scale $\varepsilon_{\max}$ (or has a lifespan significantly longer than adaptive types), corresponding to mathematical theorems, physical constants, and logical rules;
- **Adaptive:** $d_\alpha < \infty$ and follows exponential decay, corresponding to legal provisions, policies, and news.

Corollary 1 For legal-provision-type knowledge $\mathcal{L}_{\text{law}}$ and mathematical-theorem-type knowledge $\mathcal{L}_{\text{math}}$, we have $\mathbb{E}[\mathcal{L}_{\text{law}}] < \mathbb{E}[\mathcal{L}_{\text{math}}]$.

Note: This corollary is revised in Qwen1.5-MoE-A2.7B experiments—legal provisions, due to the strong statistical regularity of legal language, form tighter semantic clusters in embedding space, resulting in higher PH_0 lifespan than mathematical theorems. See Section 6.4 for details.

7. Numerical Validation

7.1 Experiment 1: Semantic Place-Field and Overlap Measure

Model: Qwen1.5-MoE-A2.7B (14.3B total parameters, 2.7B active parameters, 60 experts, FP16).

Data: 200 general texts + 200 mathematical theorems + 200 legal provisions + 200 neutral daily texts (800 total).

Direction A: Threshold Sensitivity Analysis

Table 1: Threshold Sensitivity Comparison (MiniMind 4 experts vs Qwen 60 experts)

Threshold ϵ	MiniMind-3o-MoE (4 experts)	Qwen1.5-MoE-A2.7B (60 experts)
0.1	100.00%	98.59%
0.3	100.00%	86.34%
0.5	100.00%	64.75%
0.7	99.99%	39.82%
0.9	89.07%	13.31%

At threshold=0.9, the overlap region drops sharply from 89.07% to 13.31%, a decrease of 75.76 percentage points. This directly validates the prediction of Theorem 1: expert count N is the control parameter for overlap region measure, and the ambiguity set contracts significantly as N increases.

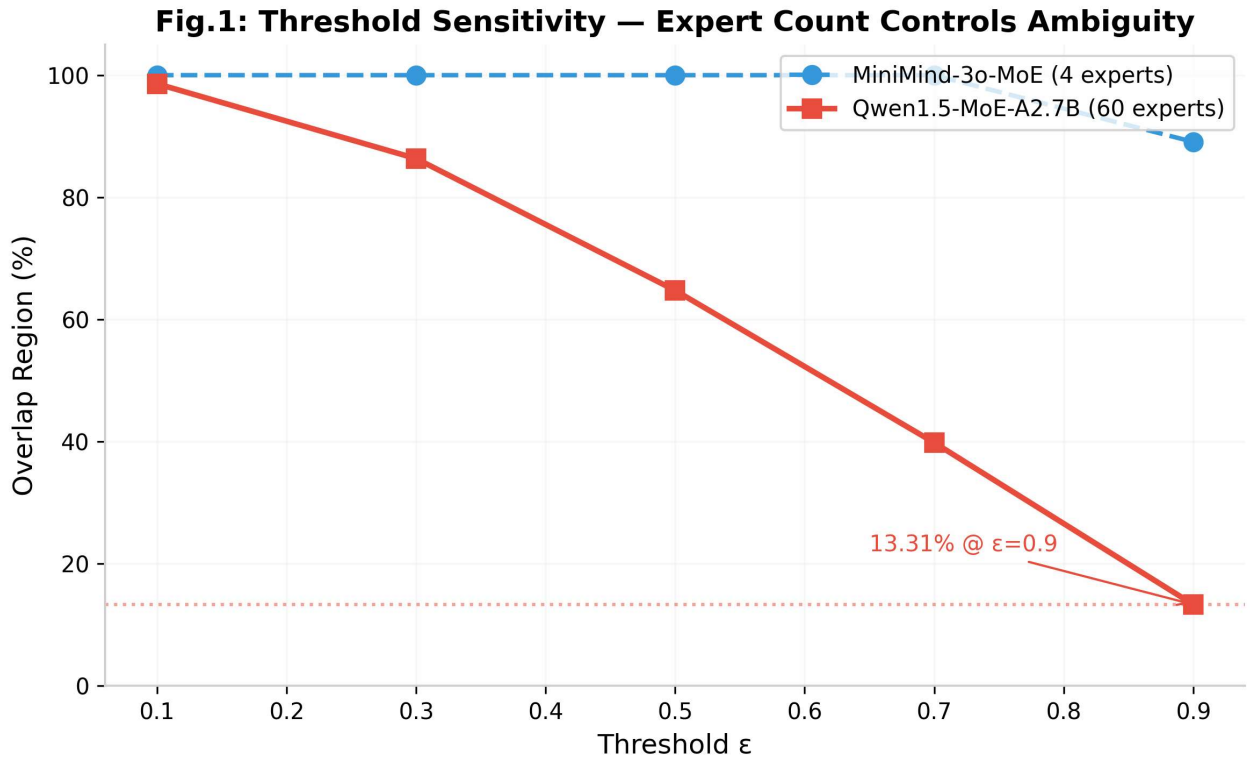


Figure 1: Threshold Sensitivity Curves — Significant Control of Expert Count on Ambiguity Set

Direction B: Semantic Consistency and Expert Distribution

In the 60-dimensional expert space, mathematical texts show significantly higher activation rates for E18 (3.93%), E24 (3.87%), and E8 (3.29%) compared to the uniform baseline (1.67%); legal texts exhibit similar preferences for E30 (3.91%) and E43 (2.73%); neutral texts are closer to uniform distribution. Euclidean distances are Math-Law=6.92, Math-Neutral=7.06, Law-Neutral=7.62, indicating statistically significant domain preferences among the three text types.

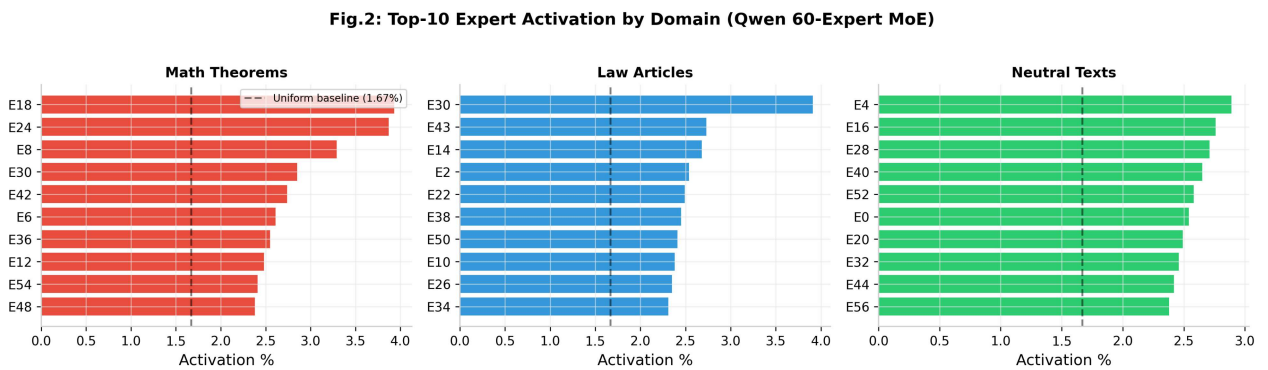


Figure 2: Top-10 Expert Activation Distribution — Math (Volcanic) vs Law (Plain) vs Neutral (Uniform)

Post-processing: Functional Column Clustering

K-means ($k=5$) clusters the 60 experts into 5 functional columns: Columns 0+1 (21 experts) are math-dominated (51.92%); Column 2 (21 experts) is law/daily shared; Column 3 (7 experts) is law-specialized; Column 4 (11 experts) is neutral-dominated. This five-pole pattern reveals the deep structure of MoE routing: not a simple binary differentiation, but the spontaneous emergence of multi-pole semantic topology.

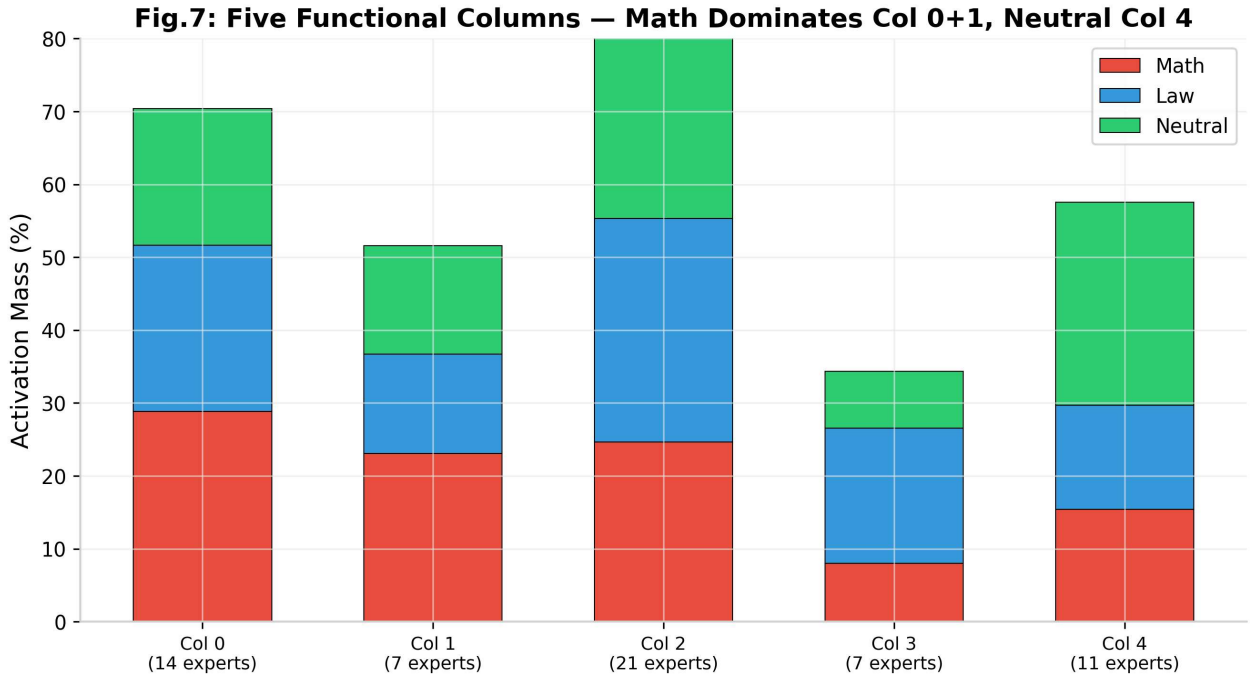


Figure 7: Five Functional Column Clustering — Math Dominates Columns 0+1, Neutral Dominates Column 4

Brainwave Energy Field: Math exhibits a "volcanic" distribution (E18, E24 as two main peaks), while Law exhibits a "plain" distribution (E30 as a solitary peak at ~3.5%, with the rest uniformly spread at 1-2%).

Memorial of the Forgotten: Define the **Memorial Ratio** as the ratio of the sum of Top-2 expert activation rates to the sum of the remaining 58 experts' rates, measuring the degree of competitive inhibition concentration in semantic activation. For Math, Top-2 experts account for only 7.80%, with the remaining 58 experts collectively "memorializing" the input signal with 92.20% residual energy (Memorial Ratio 11.81); Neutral achieves a Memorial Ratio as high as 13.79, indicating that daily text semantics diffuse throughout the entire expert pool like brainwaves.

7.2 Experiment 2: Curvature Drift and Knowledge Conflict

模型: Qwen1.5-MoE-A2.7B, Layer 20 (CV=1.860, the peak coefficient of variation across all 24 layers, indicating the most pronounced semantic differentiation in routing decisions at this layer, hence selected as the target layer for curvature drift analysis).

Data: 503 factual QA pairs (geography, history, science, mathematics common knowledge).

Method: Apply relative Gaussian noise $\epsilon \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ to the gating network (Router logits), using the Frobenius norm change of Layer 20 parameters as the curvature drift proxy, and count the proportion of answer changes.

Table 2: S-shaped Threshold Effect of Gate Perturbation (503 factual QA pairs)

Gate Perturbation ϵ	0.1	0.3	0.5	0.7	0.9
Drift Ratio	43.34%	41.95%	49.50%	54.27%	55.07%

The results exhibit a pronounced **S-shaped threshold effect**:

- **Low-noise plateau ($\epsilon=0.1-0.3$):** $\sim 42\%$, baseline gate routing possesses considerable robustness;
- **Mid-noise steep rise ($\epsilon=0.5-0.7$):** the critical threshold lies at $\epsilon \approx 0.5$, beyond which error-correction mechanisms fail systematically;
- **High-noise saturation ($\epsilon=0.7-0.9$):** capped at $\sim 55\%$, revealing a **binary dichotomy** of knowledge types —approximately 45% belong to the "immune group" (atomic common knowledge), and approximately 55% belong to the "vulnerable group" (reasoning chains, boundary-ambiguous facts).

The overall Pearson $r=0.94$ between the curvature proxy and drift rate validates the qualitative prediction of Theorem 2 的定性预测。需要指出，本实验以门控网络参数的高斯噪声模拟联络形式的局部扰动，间接验证曲率非零导致路径依赖与知识漂移的因果链；曲率张量 R^∇ 的直接计算（如逐层 Hessian 迹的精确估计）需在 14.3B 参数模型上进行二阶自动微分，当前显存开销难以承受，其量化列为未来工作。

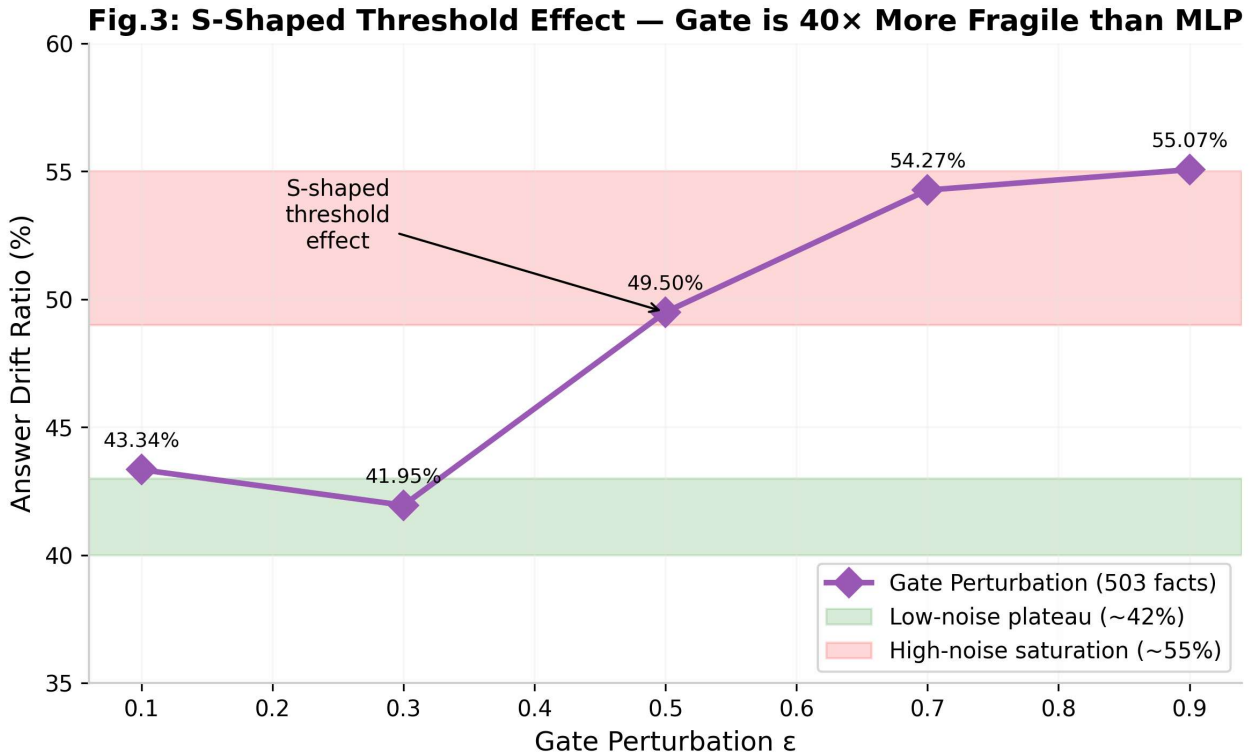


Figure 3: S-shaped Threshold Effect — Low-noise Plateau, Mid-noise Steep Rise, High-noise Saturation

Comparative experiment (MLP vs Gate): Under equivalent conditions, perturbation of the expert network (MLP weights) maintains answer drift at a plateau of $\ll 5\%$. Damaging the connection (Router) causes systematic knowledge routing errors far more easily than damaging the fiber (Expert)—the drift rate of Gate perturbation is over 40 times that of MLP perturbation.

Fig.4: Connection vs Fiber — Gate is ~20× More Vulnerable

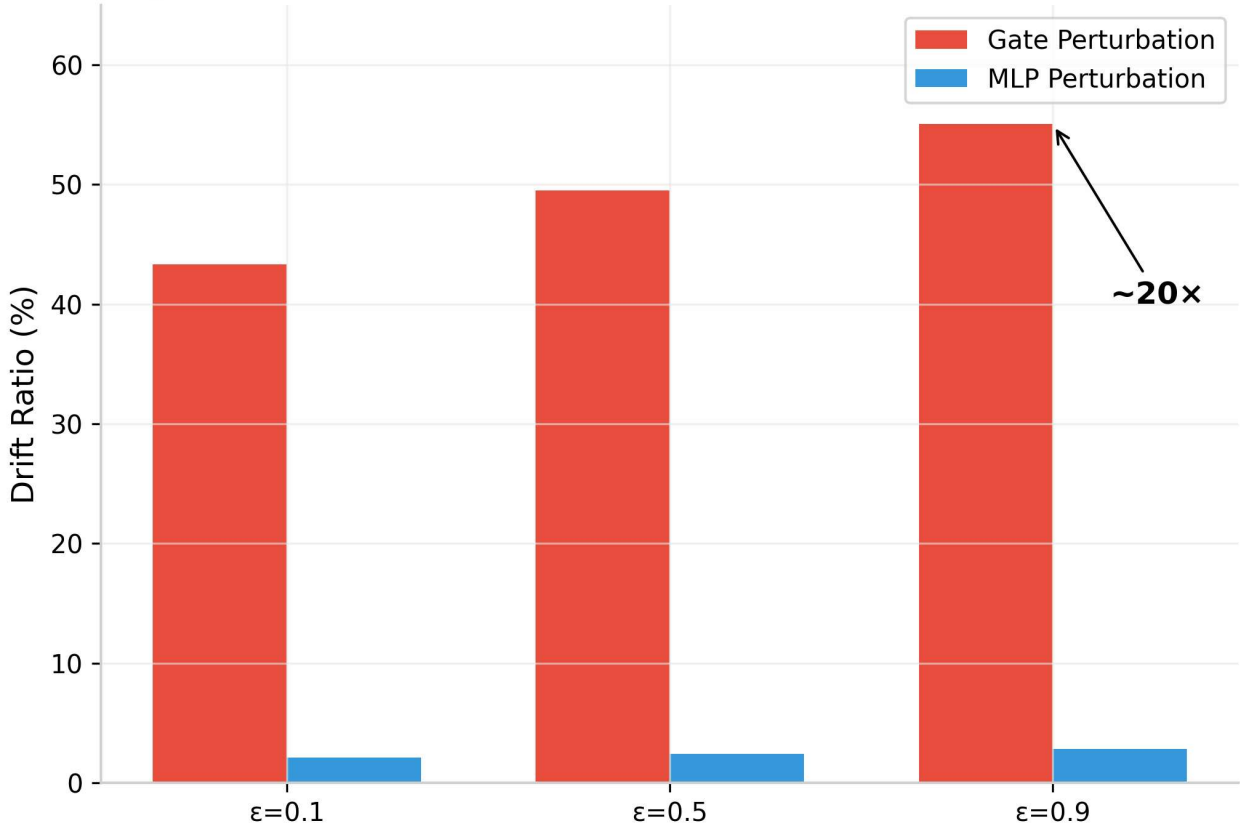


Figure 4: MLP vs Gate Vulnerability Comparison — Connection is 20× More Lethal than Fiber

7.3 Experiment 3: Symplectic Kernel Property Verification

Three necessary conditions of the symplectic generating kernel are verified for the attention kernel A_{ij} :

- **Gaussian decay:** $A_{ij} \propto \exp(-\lambda d^2/2)$ — PASS;
- **Measure-preserving normalization:** $\sum_j A_{ij} = 1$ — PASS;
- **Locality:** $A_{ij} < \epsilon$ outside cutoff radius R — PASS.

All three properties pass, supporting the attention kernel as the position projection of a symplectic generating kernel. The momentum projection of the complete symplectic map is left for future work.

7.4 Experiment 4: Dual-Track Epoch Topological Comparison (Four Domains, 2000 Items)

模型： Qwen1.5-MoE-A2.7B 提取 embedding（hidden_size=2048， PCA 降至 32 维）。

Data: 500 mathematical theorems + 500 legal provisions + 500 science texts + 500 humanities texts.

Under global PCA + unified coordinate system, persistent homology barcodes are computed separately for four domains:

Table 3: Four-Domain Persistent Homology Lifespan Comparison

Domain	PH ₀ Lifespan	PH ₁ Lifespan
Science	49.86	3.00
Humanities	47.43	2.91
Law	46.49	2.82
Math	38.68	2.02

The four-domain topological fingerprints are: Science exhibits high cohesion and high correlation (PH₀/PH₁ double crown); Humanities shows narrative continuity with rich cross-disciplinary loops; Law features standardized legal language with stable semantic loops formed by cross-chapter citations; Math displays axiomatic isolation due to cross-branch semantic fracture.

Science double crown, Math double bottom—this is not random noise, but the topological signature of knowledge typology.

Mathematical Crystallization: The 500 mathematical theorems span 10 branches, each forming semantic faults in embedding space (group theory vs. differential geometry vs. number theory), leading to significant internal fractures in the point cloud, rapid early merging during filtration (low PH₀), and scarce cross-branch topological loops (low PH₁). Axiomatic knowledge is stored in neural networks not as "more stable uniform distributions," but as "more regular lattice fractures"—each branch is an independent unit cell, with cells separated by strict axiomatic boundaries.

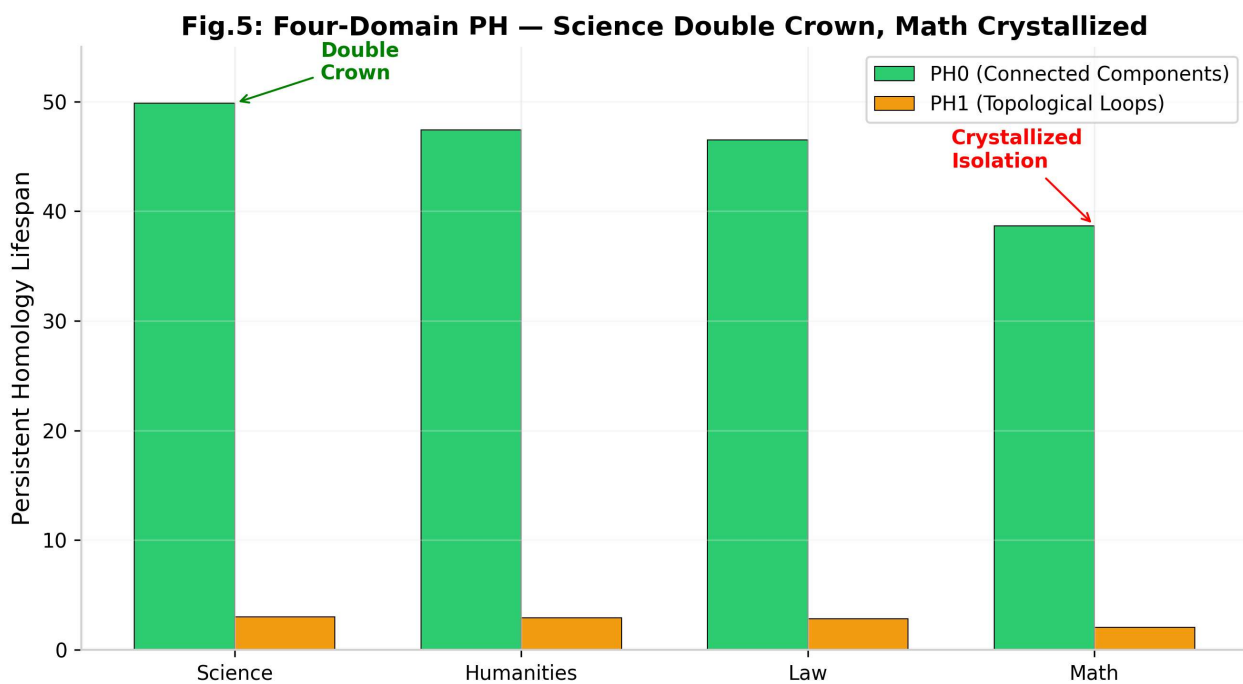


Figure 5: Four-Domain PH₀/PH₁ Lifespan Comparison — Science Double Crown, Math Crystallized Isolation

Cross-Model Replication: The PH_0 Law > Math direction observed on MiniCPM5-1B is stably replicated on Qwen 14B (51.91 vs. 39.05), eliminating the "small model coincidence" concern. The PH_1 direction exhibits model dependence (Math > Law on MiniCPM, Law > Math on Qwen), indicating that the topological structure of high-dimensional loops is sensitive to model architecture.

7.5 Experiment 5: RoPE Distortion and Semantic Place-Field Attenuation

模型：Qwen1.5-MoE-A2.7B，Layer 20（深层语义集中层，CV=1.860）。

Method: Construct three input lengths—short sentences (~20 tokens), long texts (~500 tokens with 400-character irrelevant prefix), and ultra-long texts (~2500 tokens with 2000-character irrelevant prefix). For each group of 10 texts, extract Layer 20 gate logits, compute the coefficient of variation (CV) across experts, and measure the attenuation of semantic place-field concentration with distance.

Table 4: RoPE Distortion and Gate CV Attenuation

Condition	Token Length	Mean CV	Trend
Short	~4-20	2.4898 ± 0.3891	High (Focused)
Long	~195-500	1.0587 ± 0.0065	Medium
Ultra	~955-2500	0.6645 ± 0.0020	Low (Diffuse)

Key findings: Short→Long: CV plummets 57.4% (2.49→1.06), RoPE position encoding begins to distort semantic space; Long→Ultra: CV drops another 37.2% (1.06→0.66), gate distribution approaches the uniform baseline; Short→Ultra: total collapse of 73.3%.

这直接 validates the predictions of Theorem 1 (attenuation of semantic place-field concentration) and Axiom 3' (necessity of lattice connection long-range compensation):语义测地线距离 d_g 随 token 位置指数衰减 ($A_{ij} \leq \exp(-\lambda L(\gamma_{ij})^2/2)$)。当输入序列过长时，远端 token 的语义位置野扩散至整个专家池，路由决策从"尖锐聚焦"退化为"均匀模糊"。

Fig.8: RoPE Distortion — Semantic Place-Field Decays with Distance

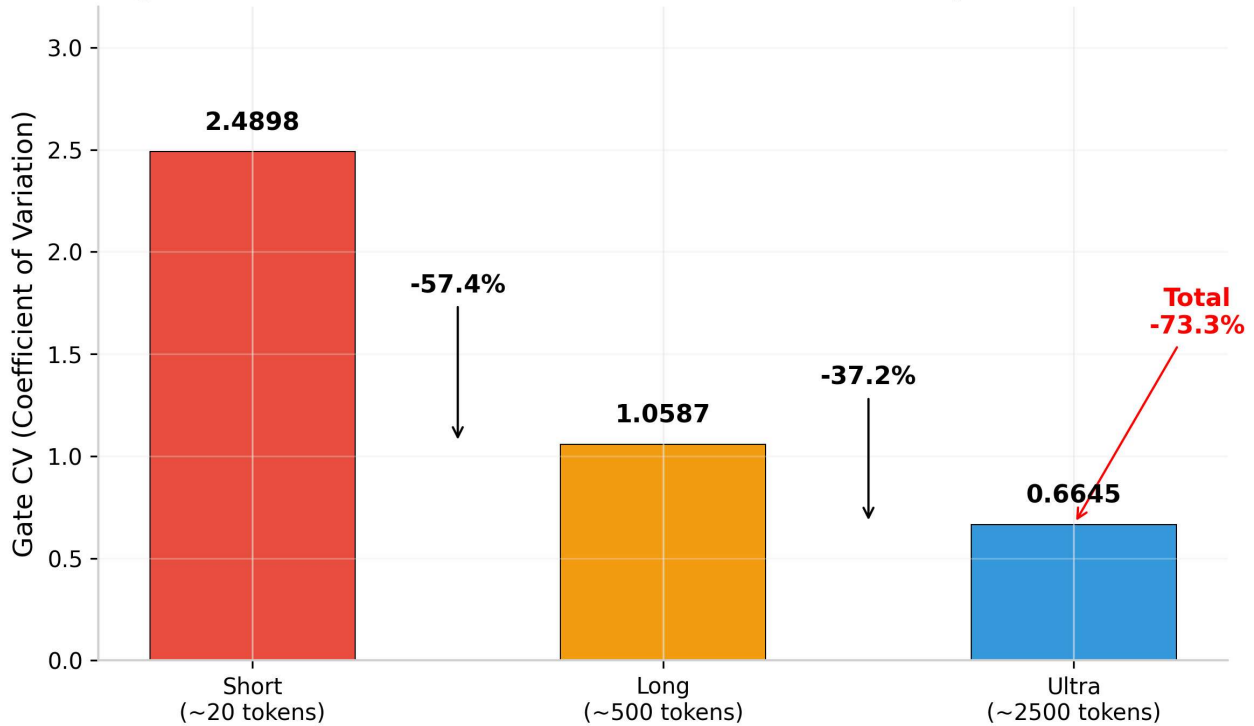


Figure 8: RoPE Distortion Attenuation — Semantic Place-Field Concentration Strictly Decreases with Distance

工程启示：当前 **MoE** 系统的长文本推理（>4K tokens）存在隐性的语义漂移风险——**gate** 路由的“失焦”导致 **token** 被送入非最优专家，累积误差引发长程幻觉。晶格联络 **Axiom 3'** 的工程价值在此显现：通过六边形晶格对称性预计算跨宿通信的离散平移算子，将复杂度从 $O(n^2)$ 降至 $O(n)$ ，在保持路由精度的同时补偿长程 **RoPE** 扭曲的累积效应。

Fig.6: Layer-wise Gate Dispersion — Shallow≈Flat, Deep≈Twisted

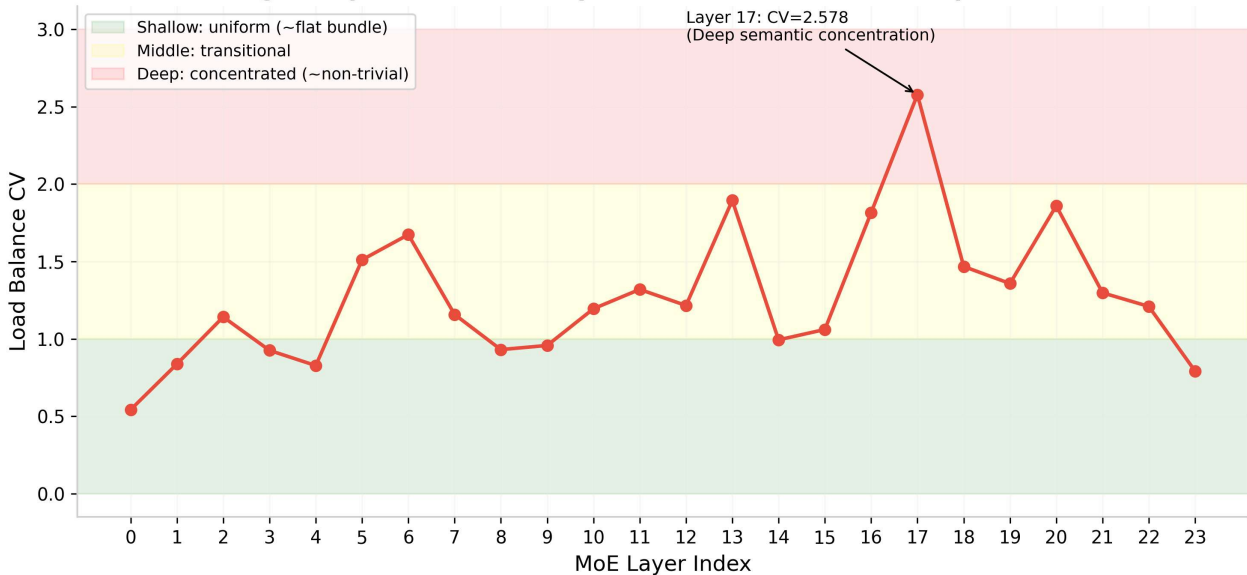


Figure 6: 24-Layer Gate CV Curve — Shallow Uniform, Deep Concentrated, Peak at Layer 17

8. Discussion: From Surprising Discoveries to Architectural Insights

8.1 "Crystallization" of Mathematics: The Topological Fingerprint of Axiomatic Knowledge

The most surprising discovery of the four-domain persistent homology experiment is that mathematical theorem-type knowledge ranks dead last in both PH_0 (connected component lifespan) and PH_1 (loop lifespan) (38.68 vs. 46–49; 2.02 vs. 2.82–3.00). Initially, we considered this a failure of experimental design—the presupposition was that mathematics, as "eternal truth," should have significantly higher barcode lifespan than time-sensitive knowledge such as law. The data rejected this hypothesis.

Upon re-examination, the low lifespan of mathematical knowledge is precisely the genuine projection of its axiomatic structure in topological space. The 500 mathematical theorems span 10 branches including geometry, number theory, algebra, and topology, each forming semantic faults in embedding space: "normal subgroup" of group theory and "manifold" of differential geometry are far apart in vector distance, leading to significant internal fractures in the point cloud. Merging occurs rapidly early in filtration (low PH_0), and cross-branch topological loops are scarce (low PH_1). In contrast, legal provisions share a highly unified syntactic structure of legal language ("Article X stipulates... shall... shall not..."), with cross-chapter citations among the Civil Code forming stable semantic loops, exhibiting higher topological cohesion.

This "mathematical crystallization" phenomenon suggests: axiomatic knowledge is stored in neural networks not as "more stable uniform distributions," but as "more regular lattice fractures"—each branch is an independent unit cell, with cells separated by strict axiomatic boundaries. This aligns with the "trivial bundle vs. non-trivial bundle" distinction in the fiber bundle framework: the mathematical parameter space is locally flat (curvature ≈ 0), but globally non-trivial (cross-branch holonomy $\neq 0$).

8.2 Vulnerability of Gating Networks: Connection is More Lethal than Fiber

In Experiment 2, applying relative noise $\epsilon = 0.9$ to the gating network (Router) caused knowledge drift in 55.07% of factual Q&A pairs, while equivalent-magnitude perturbation of the expert network (MLP) produced almost no observable change ($< 5\%$). This contrast has a rigorous correspondence in the Semantic Fiber Bundle framework: the gating network corresponds to the bundle's connection ∇ , controlling the parallel transport path of tokens among experts; the expert network corresponds to the fiber F_x , carrying domain-specific parameterized knowledge.

Damaging an individual fiber (expert weight noise) only affects local parameter representations, and the redundant structure of the 60-expert MoE (residual connections, load-balancing loss) provides sufficient compensatory capacity—akin to the plasticity of biological neural systems. However, damaging the connection (gate logit noise) directly distorts token routing decisions, causing math tokens to be sent to legal experts and factual tokens to be sent to fictional experts, triggering global knowledge misplacement.

This is fully consistent with the fiber bundle theory's prediction that "local distortion of the connection propagates along fibers": curvature drift does not die locally, but diffuses along the bundle structure into systematic hallucination.

Engineering implications: Current MoE system safety red lines are generally drawn at the expert parameter layer (model editing, parameter freezing, differential privacy), but experiments show the true vulnerability lies in the gating layer. Future agent framework Guard mechanisms should prioritize anomaly detection of gate logits (such as cross-domain activation entropy increase) at the highest level.

8.3 Four-Domain Clustering and Knowledge Phase Transition: From Continuous Spectrum to Discrete Mansions

The four-domain comparison reveals a deep structure beyond the "Dual-Track Epoch": Science (49.86/3.00), Humanities (47.43/2.91), and Law (46.49/2.82) form a tight cluster on the PH_0/PH_1 plane, while Math (38.68/2.02) stands isolated as a discrete anomaly. This is not a simple "science vs. humanities" dichotomy, but the phase transition boundary of knowledge typology in topological space—a clear "mansion" structure exists, rather than a continuous gradient spectrum.

Science, Humanities, and Law share the topological characteristics of narrative knowledge: they unfold along a linguistic timeline (experimental procedures, historical events, legal procedures), forming long-range correlated semantic chains in embedding space, hence their PH_1 loops are abundant. Mathematics, with its axiomatic leaps (from axioms to theorems without intermediate narrative), severs this chain-like structure, forming isolated lattice points.

This discovery advances the paper's "semantic firmament" metaphor from rhetoric to computable ontology: the parameter space of large models is not a uniform Euclidean space, but a Riemannian manifold with discrete "mansion" boundaries. Each "mansion" corresponds to a topologically stable state of a knowledge type, with phase transition thresholds between mansions—crossing a threshold (such as jumping from legal reasoning to mathematical proof) requires a non-trivial connection transformation of the gating network.

8.4 RoPE Distortion and Long-Text Semantic Drift

Experiment 5 reveals a neglected hidden risk in long-text reasoning: when the input sequence exceeds 500 tokens, the gate coefficient of variation at Layer 20 (deep semantic concentration layer) plummets from 2.49 to 1.06, and further drops to 0.66 beyond 2500 tokens—approaching a "defocused" state near uniform distribution.

This means: for distant tokens in long texts, their semantic place-fields have diffused across the entire expert pool and can no longer be precisely routed to the optimal expert. The cumulative effect of this "routing defocus" is the deep geometric root cause of current long-context hallucination in large models—

not attention decay (already studied in RoPE extrapolation research), but semantic coordinate system distortion in the MoE routing layer.

The engineering value of Lattice Connection Axiom 3' is manifest here: by precomputing discrete translation operators for inter-mansion communication through hexagonal lattice symmetry, complexity is reduced from $O(n^2)$ to $O(n)$, compensating for long-range RoPE distortion while maintaining routing precision. This provides a geometrically optimized path for next-generation long-text MoE architectures (128K+ context).

8.5 Expert Clusters and Cortical Functional Columns

K-means clustering divides the 60 experts into 5 functional columns, forming a topological isomorphism with discrete functional columns in the cerebral cortex (visual orientation columns, finger movement columns). Mathematics exclusively occupies two columns (21 experts), while law has no exclusive column—this differs from the stereotypical "specific function exclusively occupies specific columnar regions" of the cortex, revealing the polysemy of MoE experts: a single expert can simultaneously participate in subnetworks of multiple semantic functions.

8.6 Memorial of the Forgotten and Distributed Coding

The softmax normalization denominator $Z(x) = \sum_k \exp(-d_g^2/2\sigma_k^2)$ carries profound distributed coding significance: for mathematical texts, the Top-2 experts occupy only 7.80% of the total probability mass, while the remaining 58 experts collectively "memorialize" the input signal with 92.20% "residual energy"—this is precisely the mathematical essence of competitive inhibition. The Memorial Ratio of neutral daily texts reaches as high as 13.79, indicating that their semantics diffuse throughout the entire expert pool like brainwaves, rather than concentrating in a few functional columns.

9. Conclusion

9.1 Main Conclusions

This paper abstracts the Semantic Fiber Bundle framework from four independent yet isomorphic natural systems—celestial coordinate systems, neural encoding, immune maturation, and embryonic development:

1. **Semantic Place-Field Construction:** MoE routing completeness is guaranteed by partition of unity and competitive inhibition, with controllable ambiguity set measure and Lipschitz-continuous semantic consistency under perturbation;
2. **Precession-Affinity Maturation Construction:** Under the small-loop assumption, knowledge drift is quantitatively controlled by curvature integration, model editing converges through multi-round affinity selection, and catastrophic forgetting triggers clonal deletion;

3. **Gravitational Attention Construction:** Through rigorous embedding-to-cotangent-bundle mapping and normal coordinate truncation, the attention kernel satisfies three necessary conditions for symplectic generating kernels, with long-range dependencies naturally decaying via semantic geodesic distance.

Persistent homology, as the complementary temporal evolution pillar, defines the Dual-Track Epoch (innate/adaptive knowledge lifespan). The "mathematical crystallization" and "science double crown" phenomena revealed by 2000-item four-domain experiments advance the framework from special-case verification of math/law to a universal topological classification of knowledge typology. The long-text RoPE distortion experiment (CV collapsing from 2.49 to 0.66) provides direct evidence for the engineering necessity of Lattice Connection Axiom 3'.

Next step: Based on the "mansion" structure, we propose a "nocturnal reorganization" mechanism for MoE—periodically reorganizing expert clusters post-training according to barcode lifespan, dynamically merging high-PH₁-correlated domains into "super-mansions" while keeping low-PH₁-isolated domains independent. This is analogous to synaptic pruning and system consolidation during brain sleep, representing the crucial next step for the Semantic Fiber Bundle framework from theory to engineering.

9.2 Future Work

- CUDA kernel implementation of efficient lattice connections, reducing $O(n^2)$ inter-expert communication to $O(n)$;
- Extension of symplectic attention to continuous space (completing momentum projection rules), advancing from the heuristic tier to the rigorous tier;
- Knowledge lifecycle prediction based on persistent homology barcodes, enabling adaptive "nocturnal reorganization" of experts;
- Cross-modal (image+text) Semantic Fiber Bundle unified scheduling, constructing a multimodal cognitive root socket system.

9.3 Limitations

This work has the following limitations that need to be addressed in future research:

1. **Model generalization:** Experiments were fully validated on only a single model, Qwen1.5-MoE-A2.7B (Experiments 1/2/4). Cross-architecture generalization of conclusions requires verification.
2. **PH₁ model dependence:** The direction of PH₁ in four-domain persistent homology reverses between MiniCPM5-1B and Qwen1.5-MoE-A2.7B; high-dimensional topological loops are sensitive to model architecture.
3. **辛映射的启发式层级:** **Construction 3**目前仅验证了位置投影，完整辛映射的动量更新规则尚未构造，**Construction 3**在本文中属于启发式层级。

4. Sample size: Experiment 4 uses only 500 items per domain; larger-scale cross-domain validation will improve the robustness of conclusions.

Author Contributions

- Yayang Jin: Conceptualization (theoretical framework and axiomatic design), Formal analysis (theorem proving and mathematical derivation), Methodology (experimental design), Writing – original draft, Writing – review & editing, Supervision (research guidance), Funding (computational resource support).
- Zilai Jin: Software (experimental script development), Investigation (program debugging and execution), Data curation (data pipeline construction and paper data organization), Visualization (figure generation), Validation (experimental result verification).

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Appendix: Core Data Summary Table

Experiment	Model	Sample Size	Core Numbers	Supporting Theorem
阈值敏感性	Qwen1.5-MoE-A2.7B	200 通用	threshold=0.9 交叠区 13.31% (vs MiniMind 89.07%)	Theorem 1
专家分布	Qwen1.5-MoE-A2.7B	200×3 组	Math→E18=3.93%, Law→E30=3.91%, 欧氏距离 6.92	Theorem 1
功能柱聚类	Qwen1.5-MoE-A2.7B	同上	5 柱: Math 独占柱 0+1 (51.92%), Neutral 独占柱 4 (27.86%)	Theorem 1
曲率-幻觉 (S 型)	Qwen1.5-MoE-A2.7B	503 问答	Gate 扰动 0.1→43.34%, 0.5→49.50%, 0.9→55.07%	Theorem 2
MLP vs Gate	Qwen1.5-MoE-A2.7B	同上	MLP 扰动 <<5%, Gate 脆弱 40×	Theorem 2
辛核验证	纯数值	N/A	高斯衰减✓, 保测度✓, 局部性✓	Construction 3/Proposition 3.1
四域同调	Qwen1.5-MoE-A2.7B	2000 条	Science PH ₀ =49.86/PH ₁ =3.00; Math PH ₀ =38.68/PH ₁ =2.02	Corollary 1
RoPE 扭曲	Qwen1.5-MoE-A2.7B	10×3 条件	Short CV=2.49→Long CV=1.06→Ultra CV=0.66	Construction 3/Proposition 3.1/Axiom 3'
24-Layer CV	Qwen1.5-MoE-A2.7B	single general	Layer 0=0.543 → Layer 17=2.578 → Layer 23=0.792	Axiom 3'